UIUC airfoil coordinates database: https://m-selig.ae.illinois.edu/ads/coord_database.html

A large repository of airfoils. Most of them are formatted to be read by XFOIL/AVL.

XFOIL download: https://web.mit.edu/drela/Public/web/xfoil/

There are versions of XFOIL for both Windows and MacOS available for download.

XFOIL tutorial: XFOIL tutorial.pdf

A step-by-step tutorial for using XFOIL using many screenshots and examples.

XFLR5 link: http://www.xflr5.tech/xflr5.htm

The download link is available from here, along with some reference material. I recommend using XFLR5 for airfoil analysis because the GUI is easier to use, and it generates drag polars and L/D graphs for comparison of multiple airfoils more easily than direct XFOIL analysis. XFLR5 has 3D wing and tail analysis, and some stability calculations.

When you load an airfoil input file in XFLR5 remember to go to the "Design" drop-down menu and select "Refine Globally" and change the number of panels to 160. The default coordinates for most airfoils do not include enough data points for the analysis.

AVL download: http://web.mit.edu/drela/Public/web/avl/

There are versions of AVL for both Windows and MacOS available for download.

AVL tutorial: AVL Tutorial.pdf

A step-by-step tutorial for using AVL using many screenshots and examples, based on the example input file and airfoil file provided below.

AVL interface: AVL_Interface.exe

A windows only program developed by a former GT student that enables a visualization of the geometry while modifying. There is some capability to automate running, but it is buggy and I do not recommend using.

Example AVL input file and airfoil file: sd7062.dat and buzzed_final.avl

The example is based on an AIAA competition aircraft from ten years ago. If you are not using AVL_interface or XFLR5 to create the geometry, then it is important to start from a file that works and modify the geometry.

AirfoilDatabase and MachUpX: https://github.com/usuaero

These are Python tools that were developed by Utah State. AirfoilDatabase automates running XFOIL and generates database files that can be used in MachUpX. MachUpX is a numerical lifting line tool that integrates section properties across the span. It is much more powerful than a typical lifting line algorithm, but is still limited to AR above ~4.

OpenVSP and VSPAero: http://openvsp.org/

OpenVSP is primarily a geometry generation tool, but has several aero tools built-in. It can estimate parasite drag, wave drag, and has a built-in tool called VSPAero that uses a vortex lattice method based on an old NASA code. VSPAero is somewhat buggy and difficult to use, but the algorithm can provide some estimates for supersonic flight conditions which none of the previous tools can handle.